

Tabella 1.2 Relazioni tra i parametri delle diverse descrizioni di un doppio bipolo

| $\begin{matrix} da \rightarrow \\ a \downarrow \end{matrix}$ | $z$                                                                             | $y$                                                                                | $h$                                                                                | $\bar{m}$                                                                          | $\bar{a}$                                                                        | $\bar{b}$                                                                        |
|--------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| $z$                                                          | $\begin{matrix} z_i & z_r \\ z_f & z_o \end{matrix}$                            | $\begin{matrix} \bar{y}_o & -\bar{y}_r \\ -\bar{y}_f & \bar{y}_i \end{matrix}$     | $\begin{matrix} \bar{D}_h & \bar{h}_r \\ \bar{h}_f & 1/\bar{h}_o \end{matrix}$     | $\begin{matrix} 1/\bar{m}_i & -\bar{m}_r \\ \bar{m}_f & \bar{D}_m \end{matrix}$    | $\begin{matrix} \bar{a}_i & \bar{D}_o \\ \bar{a}_f & \bar{a}_r \end{matrix}$     | $\begin{matrix} \bar{b}_o & 1/\bar{b}_r \\ \bar{D}_b & \bar{b}_i \end{matrix}$   |
| $y$                                                          | $\begin{matrix} \bar{z}_o & -\bar{z}_r \\ -\bar{z}_f & \bar{z}_i \end{matrix}$  | $\begin{matrix} \bar{y}_i & \bar{y}_r \\ \bar{y}_f & \bar{y}_o \end{matrix}$       | $\begin{matrix} 1/\bar{h}_i & -\bar{h}_r \\ \bar{h}_f & \bar{D}_h \end{matrix}$    | $\begin{matrix} \bar{D}_m & \bar{m}_r \\ -\bar{m}_f & 1/\bar{m}_o \end{matrix}$    | $\begin{matrix} \bar{a}_o & -\bar{D}_s \\ -1/\bar{a}_r & \bar{a}_i \end{matrix}$ | $\begin{matrix} \bar{b}_i & -1/\bar{b}_r \\ -\bar{D}_b & \bar{b}_o \end{matrix}$ |
| $h$                                                          | $\begin{matrix} \bar{D}_z & \bar{z}_r \\ -\bar{z}_f & 1/\bar{z}_o \end{matrix}$ | $\begin{matrix} 1/\bar{y}_i & -\bar{y}_r \\ \bar{y}_f & \bar{D}_y \end{matrix}$    | $\begin{matrix} \bar{h}_i & \bar{h}_r \\ \bar{h}_f & \bar{h}_o \end{matrix}$       | $\begin{matrix} \bar{m}_o & -\bar{m}_r \\ -\bar{m}_f & \bar{D}_m \end{matrix}$     | $\begin{matrix} \bar{a}_r & \bar{D}_o \\ -1/\bar{a}_o & \bar{a}_f \end{matrix}$  | $\begin{matrix} \bar{b}_r & 1/\bar{b}_i \\ -\bar{D}_b & \bar{b}_f \end{matrix}$  |
| $\bar{m}$                                                    | $\begin{matrix} 1/\bar{z}_i & -\bar{z}_r \\ \bar{z}_f & \bar{D}_z \end{matrix}$ | $\begin{matrix} \bar{D}_y & \bar{y}_r \\ -\bar{y}_o & 1/\bar{y}_i \end{matrix}$    | $\begin{matrix} \bar{h}_o & -\bar{h}_r \\ -\bar{h}_f & \bar{D}_h \end{matrix}$     | $\begin{matrix} \bar{m}_i & \bar{m}_r \\ \bar{m}_f & \bar{m}_o \end{matrix}$       | $\begin{matrix} \bar{a}_f & -\bar{D}_s \\ 1/\bar{a}_i & \bar{a}_r \end{matrix}$  | $\begin{matrix} \bar{b}_f & -1/\bar{b}_o \\ \bar{D}_b & \bar{b}_r \end{matrix}$  |
| $\bar{a}$                                                    | $\begin{matrix} \bar{z}_i & \bar{D}_z \\ \bar{z}_f & \bar{z}_r \end{matrix}$    | $\begin{matrix} -\bar{y}_o & -1/\bar{y}_r \\ -\bar{D}_y & -\bar{y}_i \end{matrix}$ | $\begin{matrix} -\bar{D}_h & -\bar{h}_r \\ -\bar{h}_o & -1/\bar{h}_f \end{matrix}$ | $\begin{matrix} 1/\bar{m}_f & \bar{m}_o \\ \bar{m}_i & \bar{D}_m \end{matrix}$     | $\begin{matrix} \bar{a}_i & \bar{a}_r \\ \bar{a}_f & \bar{a}_o \end{matrix}$     | $\begin{matrix} \bar{b}_o & \bar{b}_r \\ \bar{D}_b & \bar{b}_i \end{matrix}$     |
| $\bar{b}$                                                    | $\begin{matrix} \bar{z}_o & \bar{D}_z \\ 1/\bar{z}_r & \bar{z}_i \end{matrix}$  | $\begin{matrix} -\bar{y}_i & -1/\bar{y}_r \\ -\bar{D}_y & -\bar{y}_o \end{matrix}$ | $\begin{matrix} 1/\bar{h}_r & \bar{h}_i \\ \bar{h}_o & \bar{D}_h \end{matrix}$     | $\begin{matrix} -\bar{D}_m & -\bar{m}_r \\ -\bar{m}_i & -1/\bar{m}_f \end{matrix}$ | $\begin{matrix} \bar{a}_o & \bar{a}_r \\ \bar{a}_f & \bar{a}_i \end{matrix}$     | $\begin{matrix} \bar{b}_i & \bar{b}_r \\ \bar{b}_f & \bar{b}_o \end{matrix}$     |
| $\bar{D}_z$                                                  | $\bar{z}_i \bar{z}_o - \bar{z}_r \bar{z}_f$                                     | $1/\bar{D}_y$                                                                      | $\bar{h}_i/\bar{h}_o$                                                              | $\bar{m}_o/\bar{m}_i$                                                              | $\bar{a}_r/\bar{a}_f$                                                            | $\bar{b}_r/\bar{b}_f$                                                            |
| $\bar{D}_y$                                                  | $1/\bar{D}_z$                                                                   | $\bar{y}_i \bar{y}_o - \bar{y}_r \bar{y}_f$                                        | $\bar{h}_o/\bar{h}_i$                                                              | $\bar{m}_i/\bar{m}_o$                                                              | $\bar{a}_f/\bar{a}_r$                                                            | $\bar{b}_f/\bar{b}_r$                                                            |
| $\bar{D}_h$                                                  | $\bar{z}_i/\bar{z}_o$                                                           | $\bar{y}_o/\bar{y}_i$                                                              | $\bar{h}_i \bar{h}_o - \bar{h}_r \bar{h}_f$                                        | $1/\bar{D}_m$                                                                      | $\bar{a}_i/\bar{a}_o$                                                            | $\bar{b}_o/\bar{b}_i$                                                            |
| $\bar{D}_m$                                                  | $\bar{z}_o/\bar{z}_i$                                                           | $\bar{y}_i/\bar{y}_o$                                                              | $1/\bar{D}_h$                                                                      | $\bar{m}_i \bar{m}_o - \bar{m}_r \bar{m}_f$                                        | $\bar{a}_o/\bar{a}_i$                                                            | $\bar{b}_i/\bar{b}_o$                                                            |
| $\bar{D}_s$                                                  | $\bar{z}_r/\bar{z}_f$                                                           | $\bar{y}_r/\bar{y}_f$                                                              | $-\bar{h}_r/\bar{h}_f$                                                             | $-\bar{m}_r/\bar{m}_f$                                                             | $\bar{a}_r \bar{a}_o - \bar{a}_i \bar{a}_f$                                      | $1/\bar{D}_b$                                                                    |
| $\bar{D}_b$                                                  | $\bar{z}_f/\bar{z}_r$                                                           | $\bar{y}_f/\bar{y}_r$                                                              | $-\bar{h}_f/\bar{h}_r$                                                             | $-\bar{m}_f/\bar{m}_r$                                                             | $1/\bar{D}_s$                                                                    | $\bar{b}_i \bar{b}_o - \bar{b}_r \bar{b}_f$                                      |

circuiti ci serviamo ancora delle (1-59) e a questo scopo sottraiamo e sommiamo al 2° membro della 1° la corrente  $\bar{y}_i \bar{V}_i$  e al 2° membro della 2° la corrente  $\bar{y}_r (\bar{V}_i - \bar{V}_u)$ ; si ha così:

$$I_i = (\bar{y}_i + \bar{y}_r) V_i + (-\bar{y}_r)(\bar{V}_i - \bar{V}_u), \quad I_u = (-\bar{y}_r)(\bar{V}_u - \bar{V}_i) + (\bar{y}_f - \bar{y}_i) \bar{V}_i + (\bar{y}_o + \bar{y}_r) \bar{V}_u$$

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